Laser Cutter Quick Start

Trotec Q400 RF — 60W CO_2

Good Studio — Via Faentina 244/8, Firenze

April 30, 2026

How a CO_2 Laser Cutter Works

- RF-excited CO_2 tube generates a 10.6 µm infrared beam
- Beam is steered by mirrors and focused by a lens onto the work surface
- Focused spot (∼0.1 mm) vaporizes or melts material
- **Two modes:**
  - **Engraving** — beam rasters across the surface (like a printer)
  - **Cutting** — beam traces vector paths through the material
- Air assist blows debris from the cut; exhaust extracts smoke

Key Parameters

Power	Beam intensity (%)
Speed	Head travel rate (%)
Frequency	Pulse rate (Hz)
Passes	Repeat count

What Can You Do With 60W?

Cuts well:

- Acrylic — up to ~10 mm
- MDF / HDF — up to ~6 mm
- Plywood — up to ~6 mm (depends on glue)
- Cardboard, paper, fabric
- Leather, felt, foam
- Thin wood veneers

Engraves well:

- Wood, MDF, cork
- Acrylic (frosted finish)
- Anodized aluminum
- Coated metals, glass (surface mark)
- Stone, slate, tile

Never cut:

- PVC / vinyl (emits chlorine gas)
- Polycarbonate (burns, toxic)
- ABS (toxic fumes)
- Metals (reflects beam)

Trotec Q400 RF Overview

- **Model:** Q400 RF (serial Q42-3066)
- **Tube:** 60W CeramiCore RF CO_2
- **Bed:** 1000 × 610 mm
- **Software:** Trotec Ruby (browser-based)
- **Location:** MiniPC_103 controls the laser via USB
- **Network:** `MiniPC_103` on shop Wi-Fi

Stock Sheets

Our standard stock is 1000×600 mm, 6 mm HDF.
Usable area: ∼950×550 mm (leave margin).

Tested Cut Settings (6 mm HDF)

Power	100%
Speed	0.3%
Frequency	1 000 Hz
Passes	1
Focus	Manual midplane

Safety Rules

Exhaust Fan — MANDATORY

The vent fan **must** be running before you start *any* job. Laser cutting produces toxic smoke and particulates. Running the laser without exhaust is dangerous to your health and will contaminate the workspace. **Verify airflow before every cut.**

- **Never leave a running job unattended.** Flare-ups can happen, especially with MDF and wood.
- Keep the lid closed during operation (interlock will stop the beam).
- Know where the fire extinguisher is.
- The beam is invisible (infrared) — never bypass interlocks.
- Check material safety *before* cutting. When in doubt, don't cut it.

Shop Hours

Laser use ends at **8:00 PM**. The exhaust fan is loud — please be respectful of neighbors. Plan jobs to finish before cutoff.

Step 1: Prepare Your Design

- Create an **SVG** file (Inkscape, Illustrator, Fusion, etc.)
- Use **two colors** to separate operations:

Red (`#FF0000`)
= **Cut**

Stroke color on vector paths.
Set stroke width very thin (0.01 mm).

`fill="none"`

Black (`#000000`)
= **Engrave**

Fill color for raster areas.
Text, logos, surface marks.

`stroke="none"`

Tip

Set your document size in mm to match your stock. Ruby imports at 1:1 scale when units are correct.

Step 2: Import into Ruby

1. Log into the Windows NUC:
   User: `Shamrock` / Password: `shamrock`
2. Open **Ruby** (desktop shortcut or `https://localhost:5001`)
3. Log into Ruby as admin:
   Email: `r.j.walters@gmail.com` / Password: `Shamrock1`
   (Trotec requires a registered account — just use Robb's)
4. Go to **Designs** and import your SVG
5. Double-click the design to open the **Workbench**
6. Assign the material (e.g. "HDF 6mm")
7. Ruby maps red → Cut and black → Engrave automatically

Important

If you change material settings, you must **delete the workbench** and re-create it. Ruby caches parameters at workbench creation time.

Step 3: Focus & Align

1. Place your stock on the bed
2. **Focus the lens:**
   - Use the physical gap tool (hangs on the machine)
   - Place it between the lens assembly and the material surface
   - Jog Z until the gap tool just fits — this sets focus on the surface
3. **For thick cuts ($\geq$ 6 mm):** manually raise the table $\sim$3 mm after focusing to place the focal plane at the material midplane.

   If the job also engraves, set the Engrave effect **ZOffset to +3 mm** (table back down) so engraving focuses on the surface, not the midplane.
4. **Align the job:**
   - Use Ruby's *Prepare* screen to position the design on the bed
   - The red pointer shows where the head is — use it to check corners

Focus matters

Out-of-focus cuts are wider, slower to penetrate, and produce more char. Always re-focus when changing materials or stock thickness.

Step 4: Cut

1. Verify the **exhaust fan is running**
2. Verify the **lid is closed**
3. Press **Start** in Ruby (or the physical button on the machine)
4. **Watch the job** — stay nearby the entire time
5. If you see sustained flame: press **Pause** or open the lid (kills beam)
6. You can **pause and resume** at any time — useful if you need to open the lid to adjust something, pull a finished part out while the rest is still cutting, or blow out a flare-up

After the Job

- Let the exhaust run for 1–2 minutes to clear residual smoke
- Remove your parts and clean up debris / offcuts
- Wipe the bed if there's significant char residue

Getting Good Results

- **Masking tape** on the surface reduces char marks on the face
- **Inner geometry first:** Ruby cuts holes before outlines by default (good — parts don't shift mid-cut)
- **Kerf:** the beam removes ∼0.1–0.5 mm of material. For press-fit joints, test with a kerf coupon first.
- **Taper:** with midplane focus, taper is minimal (∼0.03 mm difference top-to-bottom on 6 mm HDF). Without midplane focus, expect ∼7° taper.
- **Speed vs. quality:** slower = cleaner edges but more charring. Multi-pass at higher speed can reduce heat buildup.
- **Test first:** always run a small test piece before committing to a full sheet, especially with new materials.

Using Claude Code with the Laser

The laser can be controlled programmatically via SSH tunnels and the Ruby API.

- **Claude Code** can generate SVGs, configure materials, upload designs, and monitor jobs
- Useful for parametric designs, test sweeps, and batch operations
- The repo `trotec-tools` has the tunnel manager and Python client

Getting Started

1. Clone: `git clone github.com/project-shamrock/trotec-tools`
2. Connect: `./tunnel.sh up`
3. Open Claude Code in the repo directory
4. Ask Claude to help with your design or cutting workflow

Claude has context about the machine settings, API, material profiles, and all the lessons learned from commissioning. It can read the docs and pick up where previous sessions left off.

Questions?

Happy cutting!

Machine	Trotec Q400 RF, 60W CO_2
Software	Trotec Ruby (browser)
Materials	HDF 6 mm standard stock
Shop hours	Cutoff at 8 PM
Location	Good Studio, Via Faentina 244/8, Firenze
Repo	`project-shamrock/trotec-tools`